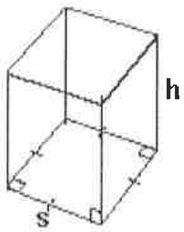
Part 2: Minimize the Surface Area of a Square Based Prism

Recall:



$$Volume\ of\ a\ Prism = (Area\ of\ Base)(Height)$$

$$Volume\ of\ Square\ Based\ Prism = (s^2)(h)$$

$$Surface\ Area\ of\ a\ Square\ Based\ Prism = 2s^2 + 4sh$$

Minimizing surface area for a given volume is important when designing packages and containers to save on materials and reduce heat loss.

Investigation: You will determine how to find the minimum surface area of a square based prism for a given volume.

a) Complete the following table to determine the dimensions of the square based prism that has a minimum surface are for a fixed volume of 64 m^3

Note: You can calculate height given the volume and side length using $h = \frac{volume}{length^2}$

	Side Length of Square Base (s)	Area of Square base (s^2)	Height (h)	Volume (s^2h)	Surface Area ($2s^2 + 4sh$)
1	1	1	64	64	258
2	2	4	16	64	136
3	3	9	7.1	64	103.2
* 4	4	16	4	64	96 *
5	5	25	2.56	64	101.2
6	6	36	1.78	64	114.72
7	7	49	1.3	64	134.4

b) Which prism has the least surface area?

Prism # 4.

c) Describe the shape of this prism

The prism is a cube. The side length of the square base = the height.

s=h.

Results:

For a square based prism with a given volume, the minimum surface area occurs when the prism is a cube.

Given a volume, you can find the dimensions of a square-based prism with minimum surface area by solving for s in the formula $V = s^3$, This is the formulas for the volume of a cube where V is the given volume and s is the length of a side of the cube. You can solve for the side length by taking the cubed root of the volume.

For example: The cubed root of 64 = $\sqrt[3]{64} = 64^{1/3} = 4$

Apply What You Learned

1) a) What would be the dimensions of the square-based prism with a minimum surface area given a volume of 27 cubic units?

$V = s^3$

$27 = s^3$

$\sqrt[3]{27} = s$

$3 = s$

The dimensions would be $3 \times 3 \times 3$.

b) Calculate the surface area ($Surface\ area\ of\ a\ cube = 6s^2$)

$SA = 6(3)^2$

$= 6(9)$

$= 54$

If a square based prism has a volume of 27units^3; the minimum surface area is 54 units^2.

2) Talia is shipping USB cables in a small cardboard square-based prism box. The box must have a capacity of 750cm^3 and Talia wants to use the minimum amount of cardboard when she ships the box.

a) What should the dimensions of the box be, to the nearest hundredth of a centimeter?

$V_{cube} = s^3$

$750 = s^3$

$\sqrt[3]{750} = s$

$9.09 = s$

The dimensions would be 9.09cm × 9.09cm × 9.09cm.

b) What is the minimum amount of cardboard that Talia will need, to the nearest tenth of a square centimeter? ($Surface\ area\ of\ a\ cube = 6s^2$)

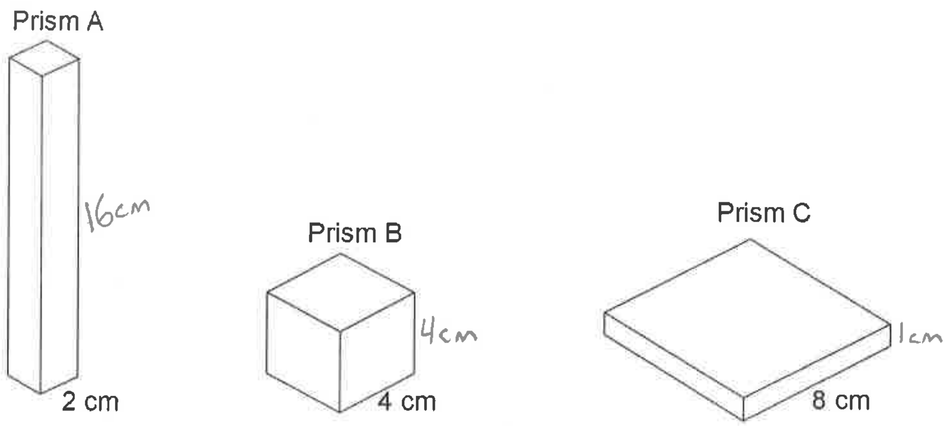
$SA = 6s^2$

$SA = 6(9.09)^2$

$SA = 495.8$

The minimum amount of cardboard needed is 495.8 cm^2.

3) Each of these square-based prisms has volume 64 cm^3.



a) Find the height of each prism.

$\boxed{h = \frac{\text{volume}}{s^2}}$

Prism A:

$h = \frac{64}{(2)^2}$

$= 16\text{ cm}$

Prism B:

$h = \frac{64}{(4)^2}$

$= 4\text{ cm}$

Prism C:

$h = \frac{64}{(8)^2}$

$= 1\text{ cm}$

b) Find the surface area of each prism.

$\boxed{SA = 2s^2 + 4sh}$

Prism A:

$SA = 2(2)^2 + 4(2)(16)$

$= 8 + 128$

$= 136\text{cm}^2$

Prism B:

$SA = 2(4)^2 + 4(4)(4)$

$= 32 + 64$

$= 96\text{cm}^2$

Prism C:

$SA = 2(8)^2 + 4(8)(1)$

$= 128 + 32$

$= 160\text{ cm}^2$

c) Order the prisms from least to greatest surface area.

B, A, C